

Quantum Index Advisors: Beyond Configuration Selection?

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Abstract

Index selection is a fundamental problem in physical database design, where the goal is to identify a set of indexes that minimizes workload cost under storage constraints. Contemporary index advisors (IA) typically adopt multi-stage pipelines comprising candidate index selection, storage calculation, benefit assignment and configuration selection. Since the problem is inherently combinatorial, a variety of heuristic pruning strategies are applied in these stages to ensure tractability. However, these heuristics may exclude beneficial configurations and result in suboptimal recommendations.

Recent quantum approaches explore using quantum computation for index selection, primarily targeting the final configuration selection stage. They formulate selection as a constrained optimization problem and leverage quantum properties such as *superposition*, *entanglement*, and *tunneling* to identify lower-cost configurations than classical IA. However, they still operate over a restricted search space shaped by the upstream IA stages.

We examine here the entire IA pipeline from a quantum perspective. Our preliminary observations suggest that a substantive gap may exist between configurations explored by current quantum pipelines, and those with lowest cost. This suggests limitations in how candidate indexes and configurations are generated, pruned, and evaluated prior to the final selection stage. Motivated by this finding, we outline a research agenda to extend quantum techniques to the entire pipeline and thereby materially improve recommendation quality.

CCS Concepts

• **Information systems** → **Autonomous database administration**; • **Hardware** → **Quantum technologies**.

Keywords

Relational Index Advisor; Quantum Computing

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1 Introduction

Index selection is a fundamental problem in physical database design of relational databases [2]. Here, the goal is to identify a set of indexes that minimizes the query workload cost while maintaining storage constraints. For this purpose, contemporary index advisors (IA), such as those in DB2 [12] and SQL Server [4], adopt multi-stage pipelines, as illustrated in Figure 1. The stages comprise (i) candidate index selection, (ii) storage calculation, (iii) benefit assignment using the query optimizer, and (iv) final configuration selection under storage constraints. To ensure tractability, a variety of heuristics are applied in these stages to reduce computational complexity. The choice of heuristics influences the shape of the explored configuration space, the candidates considered within this space, and the estimation of their benefits.



Figure 1: Typical IA Pipeline

The first stage of candidate generation extracts indexable attributes from *filter*, *join*, *group by*, and *order by* clauses. It then constructs candidate indexes such as one-column and multi-column combinations. In practice, modern systems refine this process using workload-driven heuristics, including per-query candidate extraction followed by merging into composite indexes [3]. Since the number of potential candidates may still be large, these heuristics aggressively reduce the candidate space based on estimated benefits and structural patterns, prematurely discarding candidates before downstream evaluation.

Given these index candidates, the next stage computes per-index storage overheads, which are used to guide configuration selection and identify feasible configurations under storage constraints.

The third stage of index benefit assignment introduces further approximations. Since explicitly evaluating an exponential number of index configurations is infeasible, the explored search space is restricted using heuristics such as greedy configuration construction or sampling. Benefit estimates are then derived from optimizer-estimated costs over this restricted set and attributed to individual indexes using additional heuristics [3]. These approaches approximate the configuration-level behavior and may therefore not fully capture interactions between indexes.

Finally, the last configuration selection stage identifies the best configuration among the explored candidates. In most database engines [4, 12], it is formulated as a variant of the 0/1 knapsack

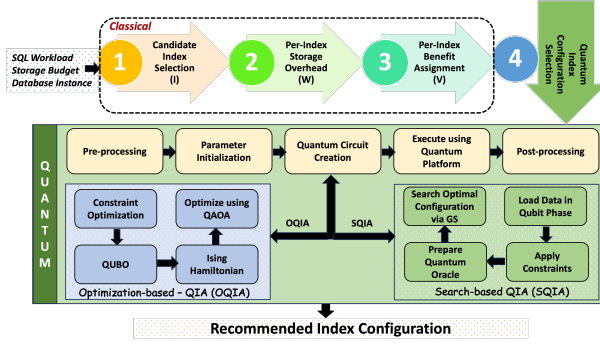


Figure 2: Quantum Index Advisor (QIA) architecture [9]

problem, where indexes are selected based on their estimated benefits under storage constraints. These formulations typically rely on additive benefit assumptions, treating index contributions as independent, and are solved using greedy heuristics to ensure efficiency. These heuristics introduce an additional layer of approximation, compounding earlier modeling assumptions and affecting the quality of the resulting index configuration.

Quantum Index Advisors

Recent research has investigated the use of quantum computation to address the combinatorial complexity of index selection [2, 9, 11]. In [11], quantum annealing was used for database index selection by formulating the configuration selection problem as a Quadratic Unconstrained Binary Optimization (QUBO) instance. Here, candidate indexes are associated with estimated workload benefits and storage costs, and storage constraints are incorporated into the objective via penalty terms. The resulting QUBO is then solved using quantum annealing hardware to identify promising configurations. Embedding strategies to map the problem onto annealing-based D-Wave hardware are also introduced [1]. This work demonstrates the feasibility of mapping index selection to a quantum annealing platform, while operating within the traditional IA pipeline.

In QIA [9], illustrated in Figure 2, two quantum IA approaches for index configuration selection were proposed. First, in the *Optimization-based Quantum Index Advisor (OQIA)*, the selection problem was formulated as a constrained optimization task encoded as a QUBO instance, similar to [11]. A difference is that this formulation is solved using the Quantum Approximate Optimization Algorithm (QAOA) [5]. QAOA is a hybrid quantum-classical method that iteratively refines a parameterized quantum circuit using classical feedback mechanisms – it is used here to carry out index selection on *gate-based* quantum computers [8].

Alternatively, in the *Search-based Quantum Index Advisor (SQIA)* approach, a Grover-style quantum search [6] was leveraged over the configuration space. In this approach, a problem-specific *quantum oracle* operates on a superposition of an exponential number of index configurations, encoded as computational basis states of a quantum system. The oracle encodes configuration specific quantities, such as storage overheads, workload benefits, and feasibility under storage constraints, using *phase based encoding* through controlled phase rotations. Finally, Grover-style amplitude amplification is

applied to increase the probability of observing low cost index configurations that satisfy the storage constraint during measurement, yielding a *quadratic* speedup over exhaustive search.

Finally, in [1], the reinforcement learning-based index advisor proposed in [10] had its conventional ϵ -greedy exploration strategy replaced with Grover search.

Motivation

The above designs highlight a dependence of quantum IA approaches on the search space induced by the initial stages of the classical IA pipeline. Since heuristic decisions are applied prior to the selection stage, the configurations considered in the final quantum selection stage are already drawn from a restricted space shaped by candidate pruning, cost modeling assumptions, and benefit estimation strategies. This raises the following question:

Can broader integration of quantum computation through the entire index advisor pipeline lead to materially better index configurations?

To investigate the above question, our preliminary observations (presented in Section 2) indicate that configurations outside those considered by existing pipelines may achieve substantially lower workload execution cost. This suggests that the limitations of current approaches may not lie solely in the optimization stage, but also in the prior upstream stages. Motivated by this observation, we outline directions for extending quantum IA techniques to the entire IA pipeline, with the expectation of identifying appreciably better index configurations.

2 Is there a Quality Gap?

To examine whether there is potential to improve on the index configurations selected by quantum IA, consider the sample IA problem instance shown in Figure 3a over a 1GB TPC-H database on the IBM DB2 platform. The workload consists of two aggregation queries over the LINEITEM table, with skewed query frequencies of 1 and 2, thereby capturing a simple yet non-uniform access pattern.

For this environment, an *exhaustive enumeration* of all 2963 feasible (i.e. within storage budget of 300 MB) one-column and two-column index configurations is presented in Figure 3b, which plots estimated workload cost against index storage consumption. The estimated costs are given in “timerons”, a DB2-defined unit for measuring abstract time. The green marker denotes the configuration selected by the quantum index advisor (QIA) [9], obtained on a 156-qubit IBM Quantum Heron processor. The shaded region highlights 2708 configurations that achieve lower estimated workload cost than QIA. Among these, the best one-column and two-column configurations, denoted by I_1 and I_2 respectively, are explicitly marked.

The QIA-recommended configuration corresponds to a multi-column index on [L_DISCOUNT | L_QUANTITY | L_TAX | L_EXTENDEDPRICE], whereas I_1 contains three one-column indexes on [L_DISCOUNT; L_QUANTITY; L_COMMITDATE] and I_2 includes three two-column indexes on [L_COMMITDATE | L_DISCOUNT; L_DISCOUNT | L_QUANTITY; L_DISCOUNT | L_TAX].

A key observation from this enumeration is that the QIA-selected configuration lies substantially above the lower envelope of the

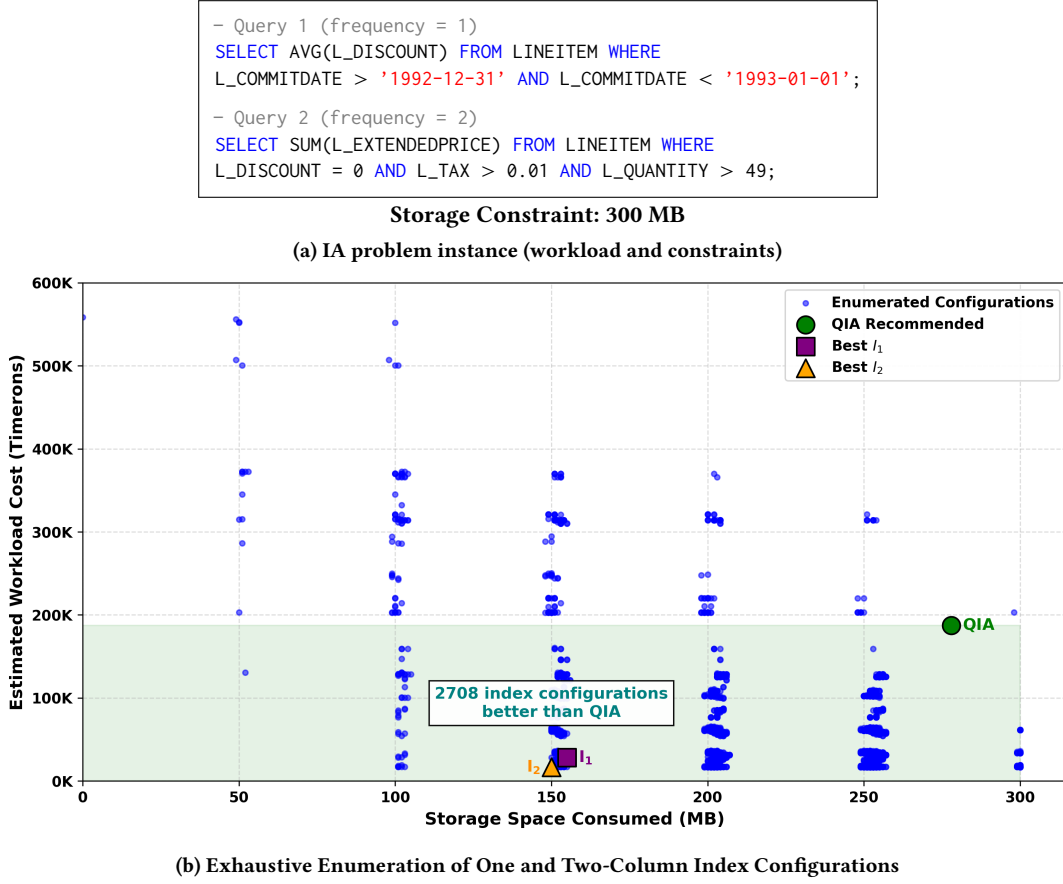


Figure 3: Sample IA problem instance and exhaustive One and Two-Column configuration landscape

configuration space, indicating that a substantial number of configurations achieve lower estimated workload cost. Specifically, the no-index baseline incurs 560K timerons, while the QIA-recommended configuration reduces this cost to 187K timerons, corresponding to a 66% reduction. In contrast, I_1 and I_2 achieve workload costs of 37K and 34K timerons, respectively, each representing approximately a 93% reduction relative to the no-index baseline, and yielding nearly a *five-fold improvement over the QIA solution*.

Apart from the quality improvement, both I_1 and I_2 represent *structurally simpler configurations* drawn from significantly smaller design spaces. While I_1 comprises only one-column indexes and I_2 is restricted to up to two-column combinations, both achieve substantially lower workload cost than the multi-column configuration identified by the quantum procedure. Notably, both configurations incur lower storage overhead than the QIA-recommended configuration, thereby providing better performance per unit storage.

These findings suggest that high-quality configurations may not be confined to complex or high-dimensional regions of the search space. Instead, effective configurations may exist even among structurally simple designs, yet remain unexplored due to restrictions imposed in earlier stages of the IA pipeline. Taken together, the observed gaps may not arise solely from limitations in the quantum selection procedure. Rather, they may also be influenced by how

the candidate space is constructed, how costs are attributed, and how configurations are presented to the optimization stage.

3 Proposed Research Agenda

In this section, we outline a set of research directions to broaden the role of quantum computation beyond configuration selection within the IA pipeline. These directions span candidate space construction, cost modeling, unified optimization formulations, and architectural co-design, collectively providing a holistic view of quantum IA.

3.1 Quantum-Aware Candidate Generation

Current candidate generation strategies rely on syntactic extraction and heuristic pruning, often discarding potentially useful index combinations before evaluation. This pruning restricts the configuration search space, which grows exponentially with the number of candidate indexes. In quantum settings, however, an exponential number of configurations can be implicitly represented in superposition. This motivates encoding the presence of candidate indexes as binary variables within a quantum optimization framework, allowing candidate generation and configuration selection to be treated jointly while exploring a broader configuration space.

However, the number of candidate indexes can be very large, and practical quantum hardware remains qubit constrained. Naively

encoding a large candidate set directly into a quantum formulation may therefore be infeasible, creating a trade off between expressiveness and hardware limitations.

A complementary direction is to employ learning-based approaches, such as LLMs, to construct a compact and semantically meaningful candidate space prior to quantum encoding. These models can augment statically generated candidates by leveraging workload semantics, query intent, and historical tuning patterns to identify high-potential index structures and eliminate redundant ones. Further, the candidate space can be adapted to the available hardware budget by controlling the number and structure of generated candidates. This may reduce the number of binary decision variables required in the quantum formulation while extending coverage of relevant configurations, thereby lowering qubit requirements while retaining expressiveness in the relevant regions of the configuration space.

3.2 Quantum-Assisted Cost Modeling

Cost estimation in existing IA pipelines often relies on decomposed benefit models that approximate configuration-level cost as additive contributions of individual indexes. While efficient, this approach may limit the ability to capture complex index interactions, where the presence of one index can influence the effectiveness of another (e.g., through join ordering, access path selection, or overlapping coverage). At the same time, directly embedding a full-fledged query optimizer cost model within a quantum circuit is likely infeasible, as it could require substantial quantum resources and deep circuit constructions. This creates a gap between simplified additive models and fully interaction-aware cost evaluation.

A promising direction could be to selectively incorporate interaction-aware corrections directly into quantum encodings. For example, limited higher-order interaction terms between indexes could be represented within QUBO formulations as quadratic penalty terms. In search-based approaches, such interaction effects could be accumulated through phase-based cost encoding without explicitly modeling the entire optimizer. This could allow the optimization procedure to reason over richer cost structures while remaining compatible with practical quantum resource constraints.

3.3 Unified Quantum Formulation

A more ambitious direction could be to treat index selection as a unified optimization problem that jointly models candidate generation, interaction-aware cost estimation, and configuration feasibility. Instead of viewing earlier IA stages as fixed preprocessing steps, their structural decisions could be expressed as constraints or penalty terms within a single optimization objective. Such a formulation could enable quantum algorithms to simultaneously explore both structural and combinatorial dimensions of the IA problem.

3.4 Quantum-Classical Co-Design

Beyond individual algorithmic extensions, a deeper opportunity may lie in rethinking index advisory as a quantum-classical co-design problem rather than a standalone optimization task. Existing approaches treat quantum computation as an accelerator applied to a fixed formulation inherited from classical IA pipelines. Such separation implicitly assumes that structural abstractions introduced in

earlier stages are sufficiently expressive. Rethinking the advisory pipeline at an architectural level may unlock configurations that remain inaccessible under current stage-separated approaches.

4 Conclusion

This paper suggests that the effectiveness of current quantum index advisors may depend not only on the final selection stage, but also on upstream IA stages. Even within the restricted space of one- and two-column configurations, exhaustive enumeration reveals a large number of lower-cost designs beyond those explored by existing quantum IA approaches. While the present study uses toy workloads due to the limits of exhaustive enumeration and current quantum hardware, these instances help isolate modeling limitations. As quantum hardware scales, broader formulations that expose more of the IA pipeline to quantum computation may become practical. For example, IBM's roadmap projects Blue Jay in 2033, with support for billion-gate circuits on 2000 qubits [7]. This points toward richer quantum IA formulations that explore larger candidate spaces and identify higher-quality index configurations.

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